

Lab Worksheet 1: Computation

CS 101 Algorithmic Problem Solving

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1. Hot Summer

Chef recently purchased a water cooler and noticed that it requires 1 hour to cool 2 liters of water.

If Chef is willing to wait for N hours after putting water in the cooler, how much water should be put in the cooler so that it is cold at the end of N hours?

Constraints

- $N \in \mathbb{Z}$
- $1 \leq N \leq 1000$

Interaction

The input comprises a single line containing the integer N .

The output must contain a single number denoting the total number of water required.

Sample

Input	Output
8	16
3	6

In the first case, $N = 8$. 16 liters of water is required to cool for 8 hours as $2 \times 8 = 16$

In the second case, $N = 3$. 6 liters of water is required to cool for 3 hours as $2 \times 3 = 6$.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
18	36
29	58
71	142

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
24	48
78	156
44	88

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT N, as 1 hour is required to cool 2 liters of water, so just multiply the hours with 2 and the answer will be how many liters of water can be cooled down. liters of water=2 x N

2. Matching Problem

There are G girl and B boy students in a school such that $B > G$.

For an upcoming sports tournament, teams of 2 are being formed such that each team comprises *exactly* 1 girl and 1 boy. Some girls may not participate.

Given G and B determine the *minimum* number of boy students who would not be able to participate.

Constraints

- $G, B \in \mathbb{N}$
- $1 \leq G < B \leq 100$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of G and B - the number of girl and boy students at the school respectively.

The output must contain a single number denoting the minimum number of boy students from the school who would not be able to participate.

Sample

Input	Output
1 3	2
3 10	7

In the first case, $(G, B) = (1, 3)$. There is only 1 girl and 3 boys. Only 1 team can be formed and, and *minimum* of 2 boys will be left behind.

In the second case, $(G, B) = (3, 10)$. There are 3 girls and 10 boys. So, maximum of 3 teams can be formed, and minimum 7 boys will be left behind.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
2 4	2
7 10	3
18 23	5

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
58, 59	1
2, 64	62
32, 97	65

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT B,G .Subtract the number of girls from the number of boys, and we will get the number of boys left behind. (LEFTOVER=B-G)

3. Ticket Fine

On a certain train, the ticket collector collects a fine of Rs. X if a passenger is traveling without a ticket. It is known that a passenger carries either a single ticket or no ticket.

There are currently P passengers on the train and they have a total of Q tickets.

Given X , P , and Q calculate the total fine collected.

Constraints

- $X, P, Q \in \mathbb{Z}$
- $1 \leq X \leq 10$
- $0 \leq Q \leq P \leq 10$

Interaction

The input comprises a single line containing 3 space-separated integers denoting the values of X , P and Q respectively.

The output must contain a single number denoting the total money collected by the ticket collector.

Sample

Input	Output
4 1 1	0
2 10 7	6

In the first case, $(X, P, Q) = (4, 1, 1)$. The total number of people traveling without a ticket are $1 - 1 = 0$. The total fine collected is $4 \times 0 = 0$.

In the second case, $(X, P, Q) = (2, 10, 7)$. The total number of passengers traveling without a ticket are $10 - 7 = 3$. The total fine collected is $3 \times 2 = 6$.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
8 5 4	8
9 7 0	63
4 5 2	12

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
4, 8, 5	12
3, 2, 0	6
4, 7, 6	4

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT P,Q,X .subtract the number of tickets from the number of passengers then people without ticket are known, multiply the number of people without tickets by the fine. FINE=(P-Q) x X

4. Mana Points

Chef is playing a mobile game. In the game, Chef's character *Chefario* can perform special attacks. Each special attack costs X mana points to *Chefario*, and she currently has a total of Y mana points.

Given X and Y determine the *maximum* number of special attacks that Chefario can perform.

Constraints

- $X, Y \in \mathbb{N}$
- $1 \leq X \leq 100$
- $1 \leq Y \leq 1000$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of X and Y respectively.

The output must contain a single number denoting the maximum number of special attacks that Chefario can perform.

Sample

Input	Output
10 30	3
6 41	6

In the first case, $(X, Y) = (10, 30)$. Chefario can perform a maximum of 3 attacks which will cost 30 mana points.

In the second case, $(X, Y) = (6, 41)$. Chefario can perform a maximum of 6 attacks which will cost 36 mana points, with 5 mana points to spare.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
50 2	0
8 68	8
9 23	2

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
34, 5	0
3, 54	18
8, 76	9

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT X,Y .divide the number of moves by the Y mana points. round down the answer and maximum number of moves can be obtained. ANSWER=Y/X round down

5. Chairs Requirement

Chef's coding class is very famous in Chefland.

This year X students joined the class and each student requires a chair to sit on. Y chairs are already present in the class.

Determine the minimum number of new chairs that Chef must arrange so that every student is able to get a chair to sit on.

Constraints

- $X, Y \in \mathbb{Z}$
- $0 \leq X, Y \leq 100$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of X and Y respectively.

The output must contain a single number denoting the minimum number of extra chairs that must be arranged.

Sample

Input	Output
20 14	6
41 41	0

In the first case, there are 20 students in the class and Chef has 14 chairs already. Therefore Chef must buy 6 more chairs.

In the second case, there are 41 students in the class and Chef already has exactly 41 chairs. Therefore Chef does not need to buy any more chairs.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
35 0	35
50 100	0
37 8	29

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
45, 5	40
88, 3	85
34, 8	26

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT=X,Y .

OUTPUT= X-Y, if answer in negative, result is zero.

6. Finding Shoes

Chef has promised to gift each of her N friends, a pair of shoes, consisting of one left shoe and one right shoe. On the way to the shoe store, Chef suddenly remembers that she already has M left shoes.

The left shoes are all right size and the shoe store allows to buy left and right shoes separately.

What is the minimum number of extra shoes that Chef will have to buy to ensure that all N friends are gifted as promised?

Constraints

- $M, N \in \mathbb{N}$
- $1 \leq N \leq 100$
- $0 \leq M \leq 100$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of N and M respectively.

The output must contain a single number denoting the minimum number of extra shoes that Chef must buy in order to ensure the gifts.

Sample

Input	Output
2 4	2
6 0	12

In the first case, $(N, M) = (2, 4)$. That is, Chef already has 4 left shoes. 2 right shoes are required form 2 pairs of shoes.

In the second case, $(N, M) = (6, 0)$. That is, Chef has no left shoes. 6 left shoes and 6 right shoes are required to form 6 pairs of shoes.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
4 3	5
8 8	8
9 3	15

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
4, 5	4
2, 6	2
6, 1	11

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

input=M,N,L

OUTPUT= $N+(N-M)$ if $N-M$ is negative, output is zero.

7. Headmaster's challenge

Your parents have enrolled you into a strange academy. The people here seem lifeless and in their eyes, you see an endless void. Nobody talks to anyone around here. Not even your dorm roommate seems to acknowledge your presence. On your first night, you say good night to your roommate before sinking into your bed after receiving no response. The next morning, you wake up in a strange room.

You look around and your attention is immediately drawn to a wall with N number of holes. In front of the wall, you see a bottomless pit with two spheres suspended above it. You also see a paper and a pen lying on the ground near the pit. As you pick up the pen and paper, you hear a voice echo in the room; it's the Headmaster.

He tells you that this is an initiation into the secret society. The academy was but a front; this is the real deal. He tells you that the pit and the spheres have a special interaction. The pit will clone the spheres if they fall into them. So if the two spheres fall, they will turn into four and be launched back up. He says the pen and paper is to write the number of times the pit is to be used before it shuts itself. So if you were to write "2" onto the paper and drop it into the pit, the two spheres will be dropped and launched back up as four, they will fall back into the pit and be launched back as eight, at which point the pit will close.

The Headmaster challenges you to determine the minimum number of uses of the pit to end up with enough spheres to fill all the N holes in the wall. He says if you end up with less spheres than required or use the pit more times than necessary, you will be discharged from the academy immediately.

Given the value of N , determine the minimum number of uses of the pit.

Constraints

- $N \in \mathbb{Z}$
- $1 \leq N \leq 2^{10}$

Interaction

The input comprises a single line containing an integer denoting the value of N .

The output must contain a single non-negative integer denoting the minimum number of uses of the pit required.

Sample

Input	Output
4	1
5	2

In the first case, $N = 4$. There are 4 holes in the wall and using the pit 1 time will give you exactly 4 spheres.

In the second case, $N = 5$. There are 5 holes in the wall, so using the pit 1 time will not give you enough spheres so you need to use the pit twice, ending up with 8 spheres.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
43	5
2^{10}	9
512	8

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
500	8
57	5
93	6

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

input N , Output = $\log_2(N) - 1$
if in decimals, round up the answer.

8. Diwali Lights

It is the eve of Diwali and Chef is decorating the house with fairy lights. The set of lights has N bulbs placed sequentially on a string which is programmed to change patterns every second.

Given N and that at least one bulb in the set is on at any given time, how many different patterns can the serial fairy lights produce?

Constraints

- $N \in \mathbb{N}$
- $1 \leq N \leq 1000$

Interaction

The input comprises a single line containing an integer denoting the value of N .

The output must contain a single number denoting the total number of patterns that can be formed.

Sample

Input	Output
1	1
2	3

Let us denote a bulb that is on as *, and a bulb that is off as -.

In the first case, $N = 1$. Only 1 pattern is possible, i.e. *.

In the second case, $N = 2$. 3 patterns are possible: -*, *- , and **.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
8	128
11	1024
4	8

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
5	16
3	4
7	64

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

input N, Output= $2^{(n)} - 1$

9. Zoo Creatures

Two types of creatures in a space zoo are very friendly to each other and must be put in the same enclosure. Type A creatures have a hands and type B creatures have b hands. They must be able to hold each others' hands in the manner described below, otherwise they emaciate and eventually vaporize.

- Each creature holds the hands of another creature type.
- Each creature should grab with all of its hands.

Given a and b , determine the smallest number of creatures of each type needed for an enclosure so that they do not end up vaporizing?

Constraints

- $a, b \in \mathbb{N}$
- $1 \leq a, b \leq 1000$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of a and b respectively.

The output must contain 2 space-separated integers denoting the minimum number of creatures of type A and type B respectively that the zoo must keep in the enclosure.

Sample

Input	Output
20 2	1 10
3 2	2 3

In the first case, $(a, b) = (20, 2)$. 1 creature of type A has 20 hands and 10 creatures of type B have a total of 20 hands. So they can grab each others' hands as required.

In the second case, $(a, b) = (3, 2)$. 2 creatures of type A have a total of 6 hands and 3 creatures of type B have a total of 6 hands. So they can grab each others' hands as required.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
1 6	6, 1
15 5	1, 3
7 3	3, 7

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
4, 7	7, 4
16, 4	1, 4
3, 5	5, 3

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT=A,B ,take the LCM of A,B,

OUTPUT= LCM/A , LCM/B